

## Initiating Search

November 11, 2025, 9:21 PM

• Search: (Diacetoxyiodo)benzene

Filtered By:

Yield: 90-100%, 80-89%, 70-79%

Document Type: Journal

## Search Tasks

| Task                                                  | Result Type                                                                                 | View                         |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------|
| Exported: Returned Reaction Results + Filters (1,182) |  Reactions | <a href="#">View Results</a> |

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

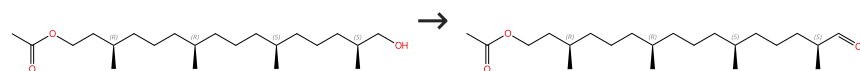
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

# Reactions (50)

[View in CAS SciFinder](#)

## Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-31129353

Steps: 1 Yield: 100%

**Enantioselective Total Synthesis of the Archaeal Lipid Parallel GDGT-0 (Isocaldarchaeol)**

By: Falk, Isaac D.; et al

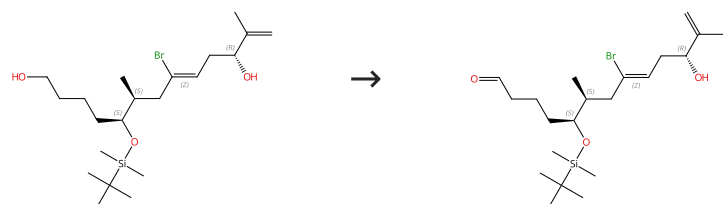
Angewandte Chemie, International Edition (2021), 60(32), 17491-17496.

1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo  
**Solvents:** Dichloromethane; 3 h, rt

Experimental Protocols

## Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Absolute stereochemistry shown,  
 Rotation (+)  
 Double bond geometry shown

 Absolute stereochemistry shown,  
 Rotation (-)  
 Double bond geometry shown

31-480-CAS-18501665

Steps: 1 Yield: 100%

**Modular total syntheses of mycolactone A/B and its [<sup>2</sup>H]-isotopologue**

By: Saint-Auret, Sarah; et al

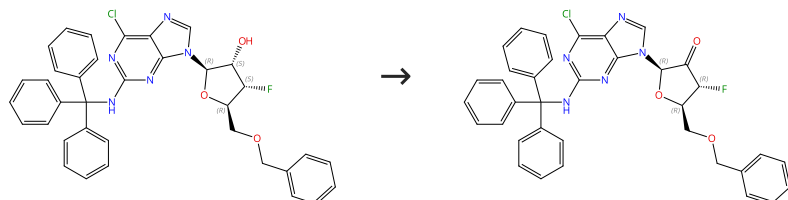
Organic &amp; Biomolecular Chemistry (2017), 15(36), 7518-7522.

1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo  
**Solvents:** Dichloromethane; 25 h, rt

1.2 **Reagents:** Sodium bicarbonate  
**Solvents:** Water

## Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-480-CAS-22635857

Steps: 1 Yield: 100%

**Synthesis and Anti-HCV Activity of Sugar-Modified Guanosine Analogues: Discovery of AL-611 as an HCV NS5B Polymerase Inhibitor for the Treatment of Chronic Hepatitis C**

By: Wang, Guangyi; et al

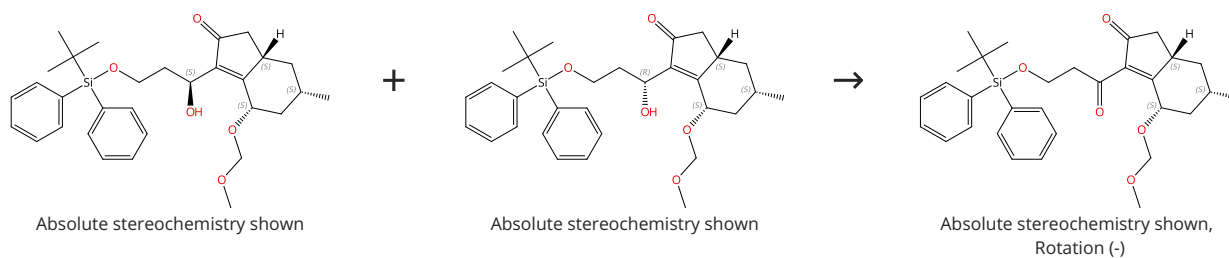
Journal of Medicinal Chemistry (2020), 63(18), 10380-10395.

1.1 **Reagents:** 2-Mercaptoethanol, Acetic acid, Iodobenzene diacetate  
**Solvents:** Dichloromethane; 0 °C; 16 h, rt

1.2 **Reagents:** Sodium thiosulfate  
**Solvents:** Dichloromethane, Water; rt

## Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-22166968

Steps: 1 Yield: 100%

**Asymmetric Total Synthesis of Fawcettimine-Type Lycopodium Alkaloid, Lycopoclavamine-A**

By: Kaneko, Hiroki; et al

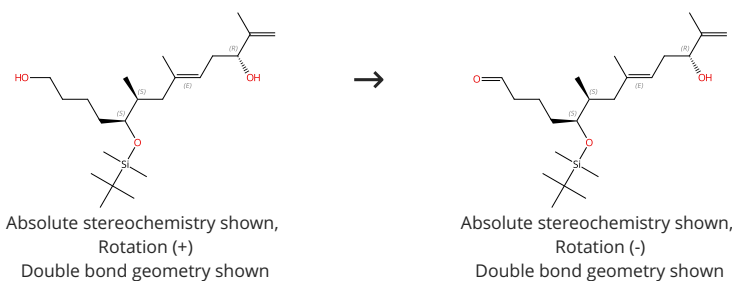
Journal of Organic Chemistry (2019), 84(9), 5645-5654.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: 2-Hydroxy-2-azatricyclo[3.3.1.1<sup>3,7</sup>]decane  
Solvents: Dichloromethane; 19.5 h, rt
- 1.2 Reagents: Sodium thiosulfate  
Solvents: Water; rt

Experimental Protocols

## Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-20243286

Steps: 1 Yield: 100%

**Synthetic strategies towards mycolactone A/B, an exotoxin secreted by Mycobacterium ulcerans**

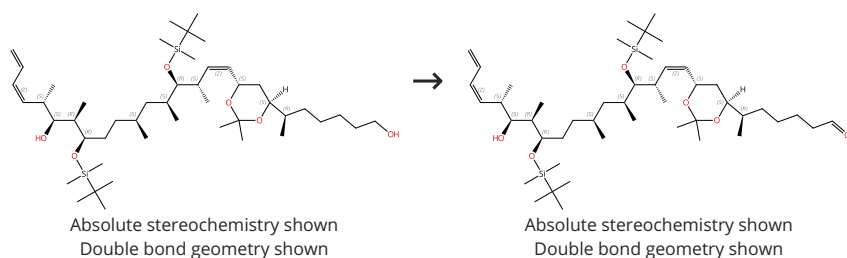
By: Saint-Auret, Sarah; et al

Organic Chemistry Frontiers (2017), 4(12), 2380-2386.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: Tempo  
Solvents: Dichloromethane; 25 h, rt
- 1.2 Reagents: Sodium bicarbonate  
Solvents: Water; rt

## Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-8176702

Steps: 1 Yield: 100%

**Total synthesis and biological evaluation of novel C2-C6 region analogues of dictyostatin**

By: Paterson, Ian; et al

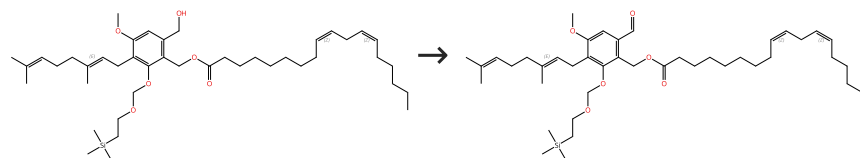
Bioorganic &amp; Medicinal Chemistry (2009), 17(6), 2282-2289.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: Tempo  
Solvents: Dichloromethane; 0 °C; 2 h, rt

Experimental Protocols

## Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

31-480-CAS-23259394

Steps: 1 Yield: 100%

**Total Synthesis, Structure Revision, and Neuroprotective Effect of Hericenones C-H and Their Derivatives**

By: Kobayashi, Shoji; et al

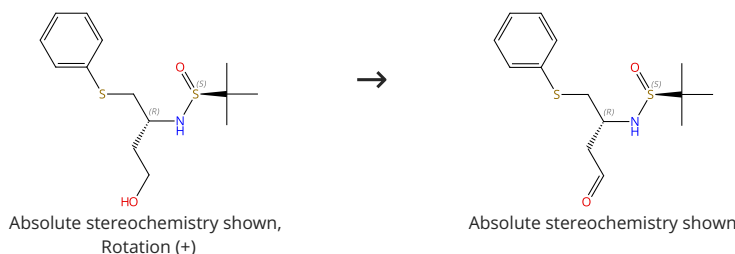
Journal of Organic Chemistry (2021), 86(3), 2602-2620.

 1.1 **Reagents:** Iodobenzene diacetate, 2-Hydroxy-2-azatricyclo [3.3.1.1<sup>3,7</sup>]decane
**Solvents:** Dichloromethane; 3 h, rt
 1.2 **Reagents:** Sodium bicarbonate, Sodium thiosulfate
**Solvents:** Water; rt

Experimental Protocols

## Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,  
Rotation (+)

Absolute stereochemistry shown

31-480-CAS-16036701

Steps: 1 Yield: 100%

**Scalable asymmetric synthesis of a key fragment of Bcl-2/Bcl-xL inhibitors**

By: Laclef, Sylvain; et al

RSC Advances (2014), 4(75), 39817-39821.

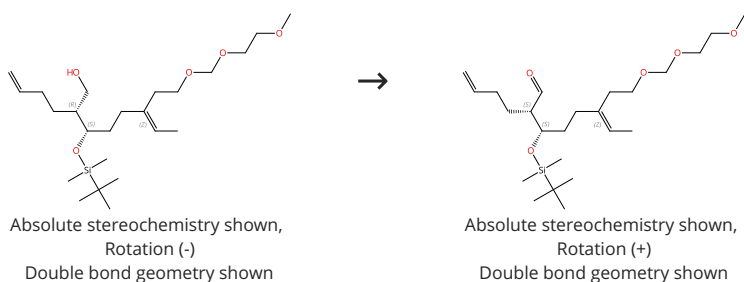
 1.1 **Reagents:** Iodobenzene diacetate
**Catalysts:** Tempo
**Solvents:** Dichloromethane; 6 h, 25 °C

 1.2 **Reagents:** Sodium sulfite
**Solvents:** Water

Experimental Protocols

## Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,  
Rotation (-)  
Double bond geometry shownAbsolute stereochemistry shown,  
Rotation (+)  
Double bond geometry shown

31-614-CAS-24036576

Steps: 1 Yield: 100%

**Asymmetric Total Synthesis of (-)-Dehydrocostus Lactone by Domino Metathesis**

By: Kaden, Felix; et al

European Journal of Organic Chemistry (2021), 2021(25), 3579-3586.

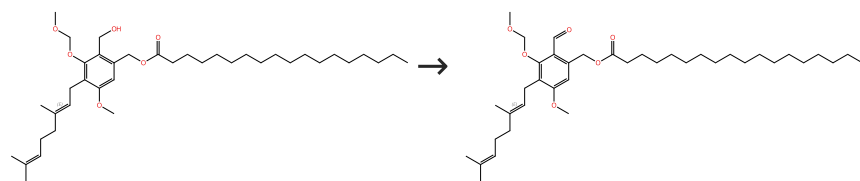
 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo  
**Solvents:** Dichloromethane; 5 h, rt

 1.2 **Reagents:** Sodium bicarbonate  
**Solvents:** Water; rt

Experimental Protocols

## Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

31-480-CAS-19425097

Steps: 1 Yield: 100%

Total syntheses and endoplasmic reticulum stress suppressive activities of hericenens A-C and their derivatives

By: Kobayashi, Shoji; et al

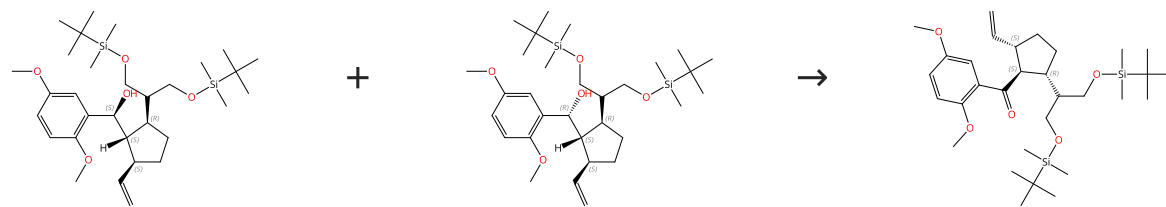
Tetrahedron Letters (2018), 59(18), 1733-1736.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: 2-Azatricyclo[3.3.1.1<sup>3,7</sup>]dec-2-yloxy  
Solvents: Dichloromethane; 3 h, rt
- 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate  
Solvents: Water; rt

Experimental Protocols

## Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

Relative stereochemistry shown

31-480-CAS-22150383

Steps: 1 Yield: 100%

Total Synthesis of Applanatumol B

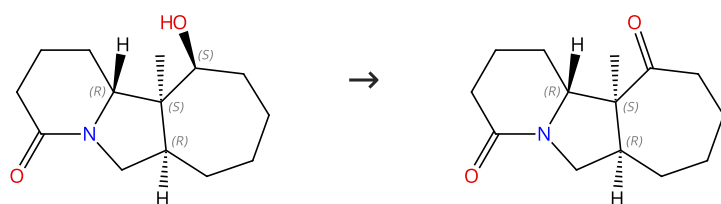
By: Uchida, Kyohei; et al

Organic Letters (2019), 21(16), 6199-6201.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: 2-Hydroxy-2-azatricyclo[3.3.1.1<sup>3,7</sup>]decane  
Solvents: Dichloromethane; 5 h, pH 7, rt
- 1.2 Reagents: Sodium thiosulfate  
Solvents: Water; rt

## Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

31-480-CAS-323055

Steps: 1 Yield: 100%

Synthetic Studies on Daphnicyclidin A: Enantiocontrolled Construction of the BCD Ring System

By: Ikeda, Shuhei; et al

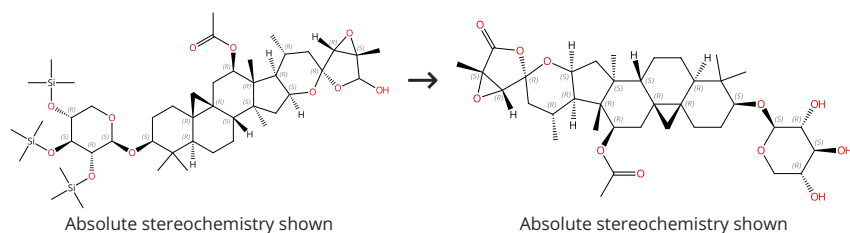
Organic Letters (2009), 11(8), 1833-1836.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: 1-Methyl-2-azatricyclo[3.3.1.1<sup>3,7</sup>]dec-2-yloxy  
Solvents: Dichloromethane; 9 h, 40 °C

Experimental Protocols

## Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-36737818

Steps: 1 Yield: 100%

Development of actein derivatives as potent anti-triple negative breast cancer agents

By: Zhang, Honglin; et al

Bioorganic &amp; Medicinal Chemistry Letters (2023), 89, 129307.

1.1 Reagents: Tempo, Iodobenzene diacetate

Solvents: Acetonitrile; 1.5 h, rt

1.2 Reagents: Sodium bicarbonate

Solvents: Water; rt

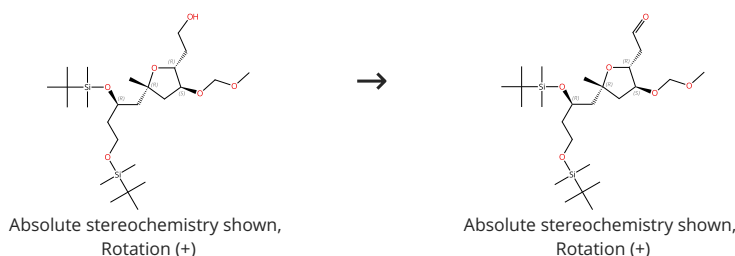
1.3 Reagents: Amberlyst 15

Solvents: Methanol, Dichloromethane; 15 min, rt

Experimental Protocols

## Scheme 14 (2 Reactions)

Steps: 1 Yield: 100%



31-614-CAS-44744340

Steps: 1 Yield: 100%

Total synthesis and structure-antifouling activity relationship of scabrolide F

By: Takamura, Hiroyoshi; et al

Organic &amp; Biomolecular Chemistry (2024), 22(28), 5739-5747.

1.1 Reagents: Iodobenzene diacetate

Catalysts: Tempo

Solvents: Dichloromethane; 0 °C; 5 h, rt

1.2 Reagents: Sodium thiosulfate

Solvents: Water

Experimental Protocols

31-614-CAS-36016949

Steps: 1 Yield: 100%

Total Synthesis of Scabrolide F

By: Takamura, Hiroyoshi; et al

Organic Letters (2022), 24(42), 7845-7849.

1.1 Reagents: Iodobenzene diacetate

Catalysts: Tempo

Solvents: Dichloromethane; 0 °C; 5 h, rt

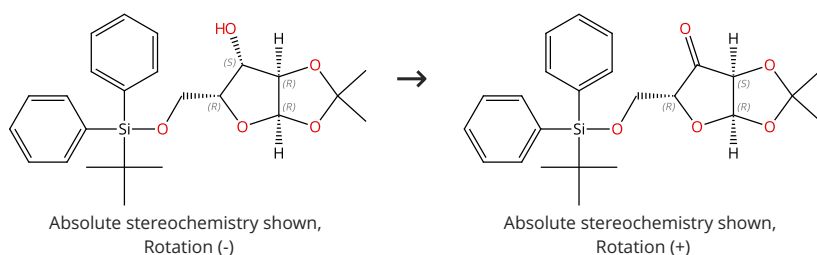
1.2 Reagents: Sodium thiosulfate

Solvents: Water; rt

Experimental Protocols

## Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



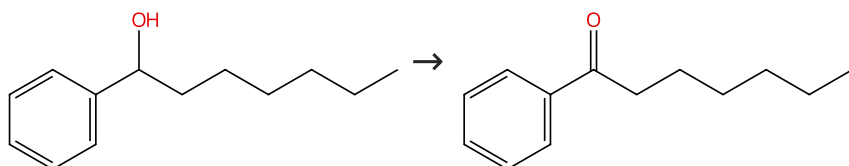
Suppliers (46)

Suppliers (46)

|                                                                                   |                      |                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-16326104                                                               | Steps: 1 Yield: 100% | <b>Synthesis of Ribonucleosidic Dimers with an Amide Linkage from D-Xylose</b><br>By: Arzel, Laurence; et al<br>Journal of Organic Chemistry (2016), 81(22), 10742-10758. |
| 1.1 Reagents: Tempo, Iodobenzene diacetate<br>Solvents: Dichloromethane; 36 h, rt |                      |                                                                                                                                                                           |
| 1.2 Reagents: Sodium sulfite<br>Solvents: Dichloromethane, Water; rt              |                      |                                                                                                                                                                           |
| Experimental Protocols                                                            |                      |                                                                                                                                                                           |

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



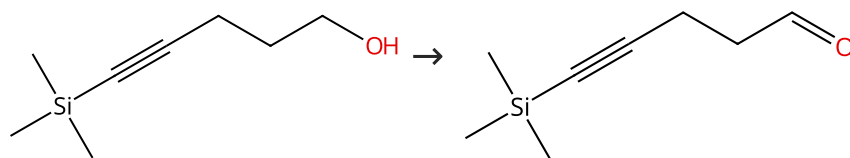
Suppliers (19)

Suppliers (60)

|                                                                                                |                      |                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-11039102                                                                            | Steps: 1 Yield: 100% | <b>Nickel-Catalyzed Mizoroki-Heck Reaction of Aryl Sulfonates and Chlorides with Electronically Unbiased Terminal Olefins: High Selectivity for Branched Products</b><br>By: Tasker, Sarah Z.; et al<br>Angewandte Chemie, International Edition (2014), 53(7), 1858-1861. |
| 1.1 Reagents: Iodobenzene diacetate<br>Catalysts: Tempo<br>Solvents: Dichloromethane; 20 h, rt |                      |                                                                                                                                                                                                                                                                            |
| 1.2 Reagents: Sodium thiosulfate<br>Solvents: Water; 15 min, rt                                |                      |                                                                                                                                                                                                                                                                            |
| Experimental Protocols                                                                         |                      |                                                                                                                                                                                                                                                                            |

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



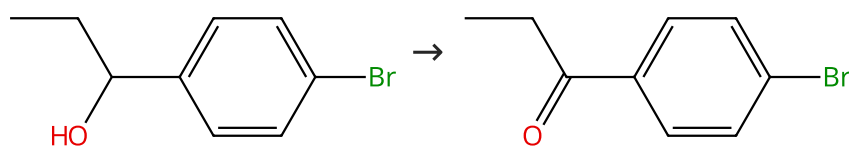
Suppliers (61)

Suppliers (6)

|                                                                                                |                      |                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-614-CAS-24036510                                                                            | Steps: 1 Yield: 100% | <b>Asymmetric Total Synthesis of (-)-Dehydrocostus Lactone by Domino Metathesis</b><br>By: Kaden, Felix; et al<br>European Journal of Organic Chemistry (2021), 2021(25), 3579-3586. |
| 1.1 Reagents: Iodobenzene diacetate<br>Catalysts: Tempo<br>Solvents: Dichloromethane; 23 h, rt |                      |                                                                                                                                                                                      |
| 1.2 Reagents: Sodium bicarbonate<br>Solvents: Water; rt                                        |                      |                                                                                                                                                                                      |
| Experimental Protocols                                                                         |                      |                                                                                                                                                                                      |

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



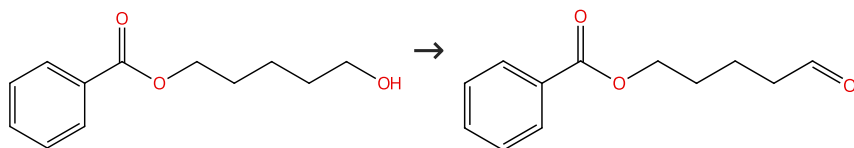
Suppliers (52)

Suppliers (63)

|                                                                                                     |                      |                                                                                 |
|-----------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------|
| 31-480-CAS-11602636                                                                                 | Steps: 1 Yield: 100% | <b>A mild and efficient oxidation of alcohols in water</b>                      |
| 1.1 Reagents: Iodobenzene diacetate<br>Catalysts: Tetrabutylammonium bromide<br>Solvents: Water; rt |                      | By: Xu, Chuan Zhi; et al<br>Chinese Chemical Letters (2004), 15(10), 1149-1152. |
| 1.2 Solvents: Water; 2 h, rt                                                                        |                      |                                                                                 |

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



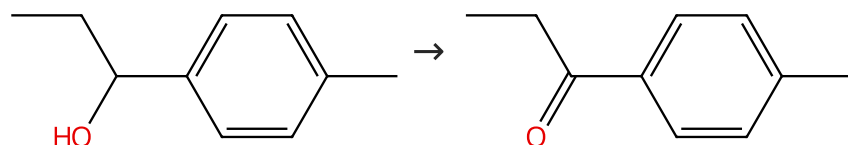
Suppliers (4)

Suppliers (3)

|                                                                                    |                      |                                                                                                |
|------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------|
| 31-480-CAS-21587194                                                                | Steps: 1 Yield: 100% | <b>Catalytic Acceptorless Dehydrogenation of Aliphatic Alcohols</b>                            |
| 1.1 Reagents: Tempo, Iodobenzene diacetate<br>Solvents: Dichloromethane; 2.5 h, rt |                      | By: Fuse, Hiromu; et al<br>Journal of the American Chemical Society (2020), 142(9), 4493-4499. |
| 1.2 Reagents: Sodium thiosulfate<br>Solvents: Water                                |                      |                                                                                                |

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



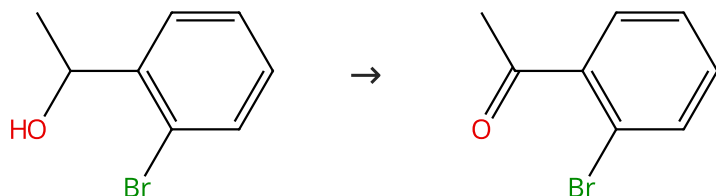
Suppliers (40)

Suppliers (74)

|                                                                                                     |                      |                                                                                 |
|-----------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------|
| 31-480-CAS-2794471                                                                                  | Steps: 1 Yield: 100% | <b>A mild and efficient oxidation of alcohols in water</b>                      |
| 1.1 Reagents: Iodobenzene diacetate<br>Catalysts: Tetrabutylammonium bromide<br>Solvents: Water; rt |                      | By: Xu, Chuan Zhi; et al<br>Chinese Chemical Letters (2004), 15(10), 1149-1152. |
| 1.2 Solvents: Water; 2 h, rt                                                                        |                      |                                                                                 |

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (60)

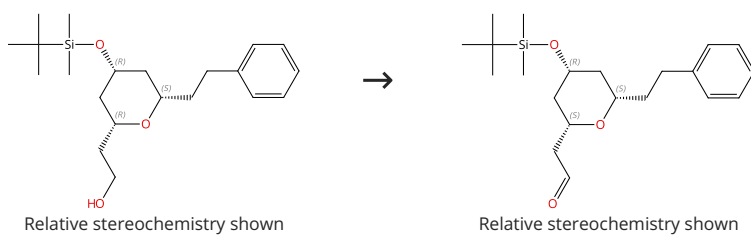
Suppliers (88)

|                                                                                                      |                      |                                                                                                                         |
|------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-16499373                                                                                  | Steps: 1 Yield: 100% | <b>Hypervalent iodine/TEMPO-mediated oxidation in flow systems: a fast and efficient protocol for alcohol oxidation</b> |
| 1.1 Reagents: Iodobenzene diacetate<br>Catalysts: Tempo<br>Solvents: Dichloromethane; 4.5 min, 35 °C |                      | By: Ambreen, Nida; et al<br>Beilstein Journal of Organic Chemistry (2013), 9, 1437-1442, 6 pp..                         |
| 1.2 Solvents: Water                                                                                  |                      |                                                                                                                         |



## Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-19714340

Steps: 1 Yield: 100%

**Oxa/thiazole-tetrahydropyran triazole-linked hybrids with selective antiproliferative activity against human tumour cells**

By: Valdomir, Guillermo; et al

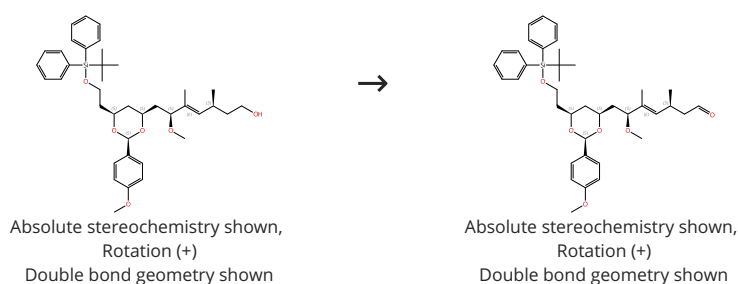
New Journal of Chemistry (2018), 42(16), 13784-13789.

1.1 **Catalysts:** Tempo, Iodobenzene diacetate  
**Solvents:** Dichloromethane; 3 h, rt

Experimental Protocols

## Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-13946895

Steps: 1 Yield: 100%

**Total Synthesis of Biselyngbyolide A**

By: Tanabe, Yurika; et al

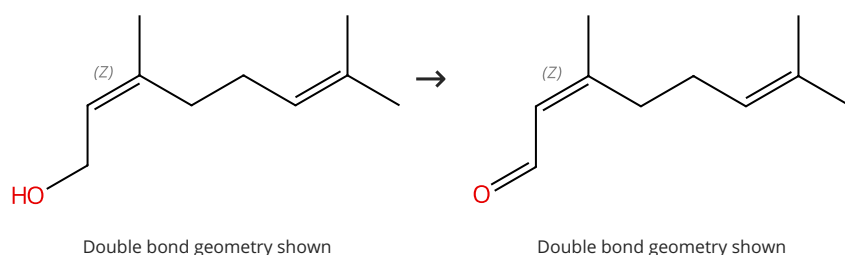
Organic Letters (2014), 16(11), 2858-2861.

1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo; 1 h, rt

Experimental Protocols

## Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (37)

31-480-CAS-4124027

Steps: 1 Yield: 100%

**Catalytic Enantioselective 1,2-Diboration of 1,3-Dienes: Versatile Reagents for Stereoselective Allylation**

By: Kliman, Laura T.; et al

Angewandte Chemie, International Edition (2012), 51(2), 521-524, S521/1-S521/178.

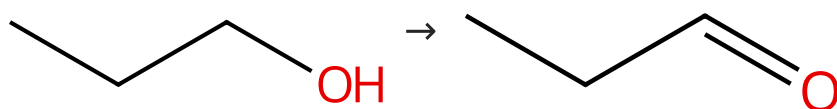
1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo  
**Solvents:** Acetonitrile, Water; 3 h, pH 7, 0 °C → rt

1.2 **Reagents:** Sodium thiosulfate  
**Solvents:** Water; rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (155)

Suppliers (75)

31-480-CAS-16499367

Steps: 1 Yield: 100%

**Hypervalent iodine/TEMPO-mediated oxidation in flow systems: a fast and efficient protocol for alcohol oxidation**

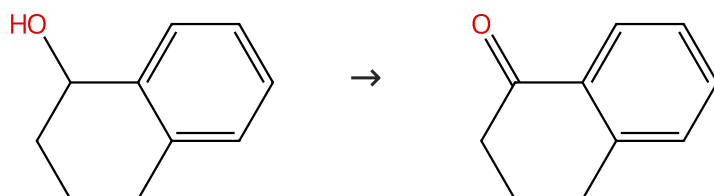
By: Ambreen, Nida; et al

Beilstein Journal of Organic Chemistry (2013), 9, 1437-1442, 6 pp..

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: Tempo  
Solvents: Dichloromethane; 4.5 min, 35 °C
- 1.2 Solvents: Water

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (98)

31-480-CAS-13397577

Steps: 1 Yield: 100%

**A mild and efficient oxidation of alcohols in water**

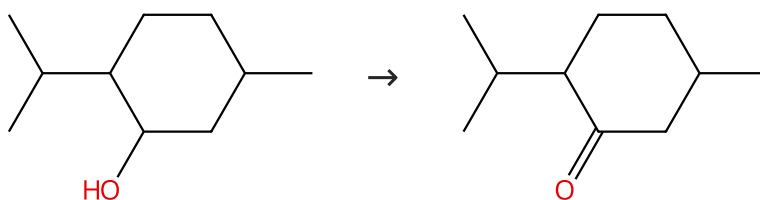
By: Xu, Chuan Zhi; et al

Chinese Chemical Letters (2004), 15(10), 1149-1152.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: Tetrabutylammonium bromide  
Solvents: Water; rt
- 1.2 Solvents: Water; 2 h, rt

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (47)

Suppliers (76)

31-480-CAS-8809396

Steps: 1 Yield: 100%

**A mild and efficient oxidation of alcohols in water**

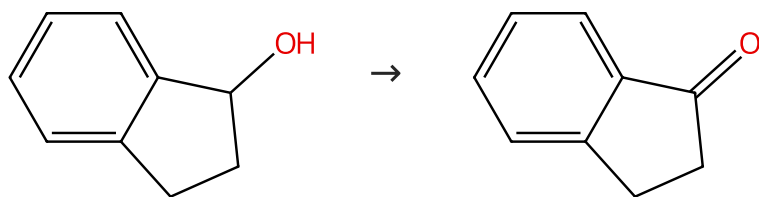
By: Xu, Chuan Zhi; et al

Chinese Chemical Letters (2004), 15(10), 1149-1152.

- 1.1 Reagents: Iodobenzene diacetate  
Catalysts: Tetrabutylammonium bromide  
Solvents: Water; rt
- 1.2 Solvents: Water; 2 h, rt

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (107)

31-480-CAS-4253382

Steps: 1 Yield: 100%

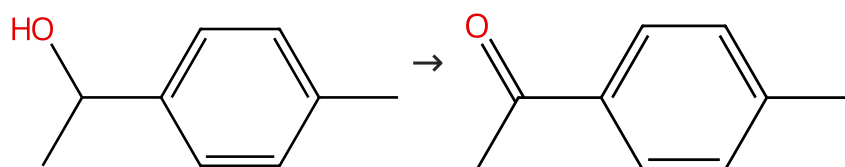
**A mild and efficient oxidation of alcohols in water**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt
- 1.2 **Solvents:** Water; 2 h, rt

By: Xu, Chuan Zhi; et al  
 Chinese Chemical Letters (2004), 15(10), 1149-1152.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (103)

31-480-CAS-355258

Steps: 1 Yield: 100%

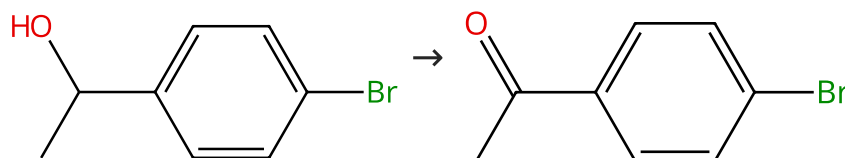
**A mild and efficient oxidation of alcohols in water**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt
- 1.2 **Solvents:** Water; 2 h, rt

By: Xu, Chuan Zhi; et al  
 Chinese Chemical Letters (2004), 15(10), 1149-1152.

Scheme 30 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (72)

Suppliers (83)

31-480-CAS-6431295

Steps: 1 Yield: 100%

**A multipolymer system for organocatalytic alcohol oxidation**

- 1.1 **Reagents:** Iodobenzene diacetate (polystyrene-supported)  
**Catalysts:** Tempo (JandaJel resin-supported)  
**Solvents:** 1,2-Dichloroethane; 12 h, 70 °C

By: But, Tracy Yuen Sze; et al  
 Organic & Biomolecular Chemistry (2005), 3(6), 970-971.

31-480-CAS-11300214

Steps: 1 Yield: 100%

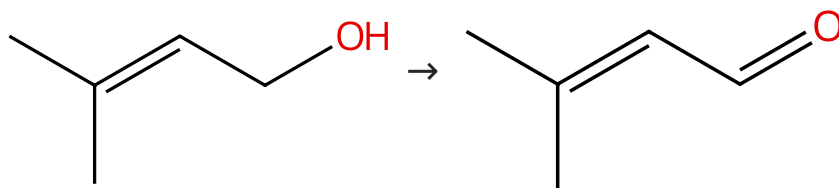
**A mild and efficient oxidation of alcohols in water**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt
- 1.2 **Solvents:** Water; 2 h, rt

By: Xu, Chuan Zhi; et al  
 Chinese Chemical Letters (2004), 15(10), 1149-1152.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (67)

31-480-CAS-15194686

Steps: 1 Yield: 100%

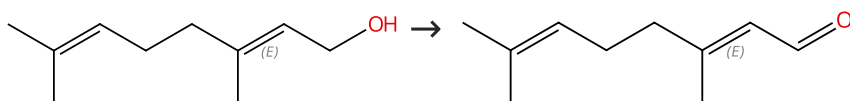
**A mild and efficient oxidation of alcohols in water**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt
- 1.2 **Solvents:** Water; 1 h, rt

By: Xu, Chuan Zhi; et al  
 Chinese Chemical Letters (2004), 15(10), 1149-1152.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Suppliers (117)

Suppliers (36)

31-614-CAS-36363401

Steps: 1 Yield: 100%

**Chemical synthesis and antifouling activity of monoterpene-furan hybrid molecules**

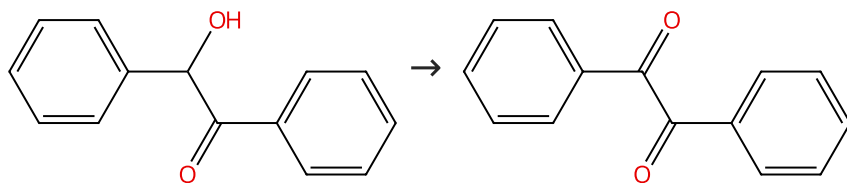
- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tempo  
**Solvents:** Dichloromethane; 14 h, rt
- 1.2 **Reagents:** Sodium thiosulfate  
**Solvents:** Water; 0 °C

By: Takamura, Hiroyoshi; et al  
 Organic & Biomolecular Chemistry (2023), 21(3), 632-638.

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (118)

Suppliers (101)

31-480-CAS-24680

Steps: 1 Yield: 100%

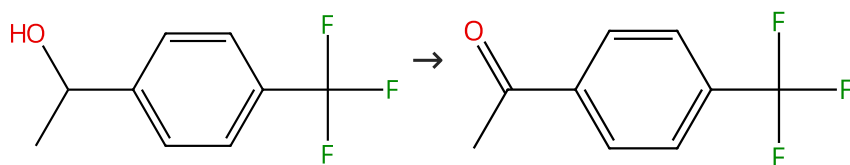
**A multipolymer system for organocatalytic alcohol oxidation**

- 1.1 **Reagents:** Iodobenzene diacetate (polystyrene-supported)  
**Catalysts:** Tempo (JandaJel resin-supported)  
**Solvents:** 1,2-Dichloroethane; 12 h, 70 °C

By: But, Tracy Yuen Sze; et al  
 Organic & Biomolecular Chemistry (2005), 3(6), 970-971.

## Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (85)

31-480-CAS-2467871

Steps: 1 Yield: 100%

**A mild and efficient oxidation of alcohols in water**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt
- 1.2 **Solvents:** Water; 2 h, rt

By: Xu, Chuan Zhi; et al  
 Chinese Chemical Letters (2004), 15(10), 1149-1152.

## Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

Suppliers (99)

31-480-CAS-7464662

Steps: 1 Yield: 100%

**Clean and highly selective oxidation of alcohols by the Ph I(OAc)<sub>2</sub>/Mn(TPP)CN/Im catalytic system**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Imidazole, (*SP*-5-31)-(Cyano-κC)[5,10,15,20-tetraphenyl-21*H*,23*H*-porphinato(2-)-κN<sup>21</sup>,κN<sup>22</sup>,κN<sup>23</sup>,κN<sup>24</sup>] manganese  
**Solvents:** Dichloromethane; 120 min, rt

By: Karimipour, Gholam Reza; et al  
 Journal of Chemical Research (2007), (4), 252-256.

## Scheme 36 (2 Reactions)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

Supplier (1)

31-480-CAS-10046369

Steps: 1 Yield: 100%

**An Expedient Entry to 9-Azabicyclo[3.3.1]nonane N-Oxyl (ABNO): Another Highly Active Organocatalyst for Oxidation of Alcohols**

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** 2-Azatricyclo[3.3.1.1<sup>3,7</sup>]dec-2-yl oxy  
**Solvents:** Dichloromethane; 14 h, rt

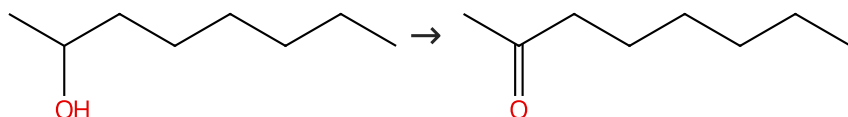
By: Shibuya, Masatoshi; et al  
 Journal of Organic Chemistry (2009), 74(12), 4619-4622.

Experimental Protocols

|                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>31-480-CAS-7764834</div> <div>Steps: 1 Yield: 100%</div> <div>           1.1 Reagents: Iodobenzene diacetate<br/>           Catalysts: 1-Methyl-2-azatricyclo[3.3.1.1<sup>3,7</sup>]dec-2-yloxy<br/>           Solvents: Dichloromethane; 9 h, rt<br/>           1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br/>           Solvents: Diethyl ether, Water; rt<br/>           Experimental Protocols         </div> | <div>2-Azaadamantane N-Oxyl (AZADO) and 1-Me-AZADO: Highly Efficient Organocatalysts for Oxidation of Alcohols</div> <div>By: Shibuya, Masatoshi; et al</div> <div>Journal of the American Chemical Society (2006), 128(26), 8412-8413.</div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 37 (3 Reactions)

Steps: 1 Yield: 100%



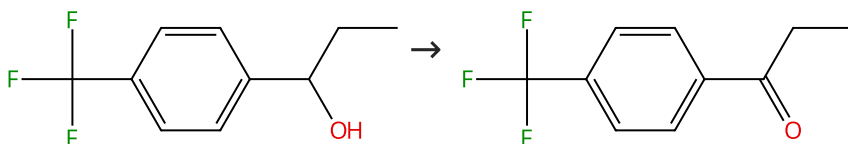
Suppliers (90)

Suppliers (86)

|                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>31-480-CAS-10945448</div> <div>Steps: 1 Yield: 100%</div> <div>           1.1 Reagents: Iodobenzene diacetate<br/>           Catalysts: Tetrabutylammonium bromide<br/>           Solvents: Water; rt<br/>           1.2 Solvents: Water; 1 h, rt         </div> | <div>A mild and efficient oxidation of alcohols in water</div> <div>By: Xu, Chuan Zhi; et al</div> <div>Chinese Chemical Letters (2004), 15(10), 1149-1152.</div>                                                     |
| <div>31-480-CAS-9268608</div> <div>Steps: 1 Yield: 100%</div> <div>           1.1 Reagents: Iodobenzene diacetate (polymer-supported)<br/>           Catalysts: Potassium bromide<br/>           Solvents: Water         </div>                                       | <div>Facile and clean oxidation of alcohols in water using hypervalent iodine(III) reagents</div> <div>By: Tohma, Hirofumi; et al</div> <div>Advanced Synthesis &amp; Catalysis (2002), 344(3+4), 328-337.</div>      |
| <div>31-480-CAS-12243155</div> <div>Steps: 1 Yield: 100%</div> <div>           1.1 Reagents: Iodobenzene diacetate (polymer-supported), Potassium bromide<br/>           Solvents: Water         </div>                                                               | <div>Facile and clean oxidation of alcohols in water using hypervalent iodine(III) reagents</div> <div>By: Tohma, Hirofumi; et al</div> <div>Angewandte Chemie, International Edition (2000), 39(7), 1306-1308.</div> |

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



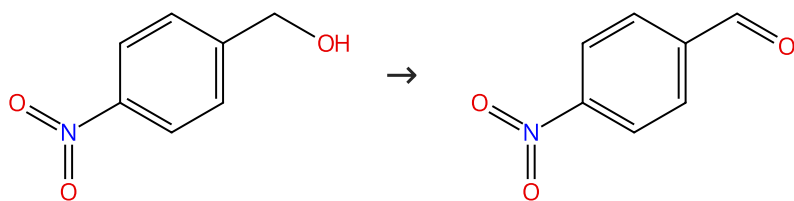
Suppliers (59)

Suppliers (73)

|                                                                                                                                                                                                                                                                      |                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>31-480-CAS-8861256</div> <div>Steps: 1 Yield: 100%</div> <div>           1.1 Reagents: Iodobenzene diacetate<br/>           Catalysts: Tetrabutylammonium bromide<br/>           Solvents: Water; rt<br/>           1.2 Solvents: Water; 2 h, rt         </div> | <div>A mild and efficient oxidation of alcohols in water</div> <div>By: Xu, Chuan Zhi; et al</div> <div>Chinese Chemical Letters (2004), 15(10), 1149-1152.</div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 39 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (95)

Suppliers (91)

31-480-CAS-1081271

Steps: 1 Yield: 100%

**Clean and highly selective oxidation of alcohols by the Ph I(OAc)<sub>2</sub>/Mn(TPP)CN/Im catalytic system**

By: Karimipour, Gholam Reza; et al

Journal of Chemical Research (2007), (4), 252-256.

- 1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Imidazole, (*SP*-5-31)-(Cyano-κC)[5,10,15,20-tetraphenyl-21*H*,23*H*-porphinato(2-)-κN<sup>21</sup>,κN<sup>22</sup>,κN<sup>23</sup>,κN<sup>24</sup>] manganese  
**Solvents:** Dichloromethane; 10 min, rt

31-480-CAS-6435575

Steps: 1 Yield: 100%

**Polymer-supported hypervalent iodine reagents in 'clean' organic synthesis with potential application in combinatorial chemistry**

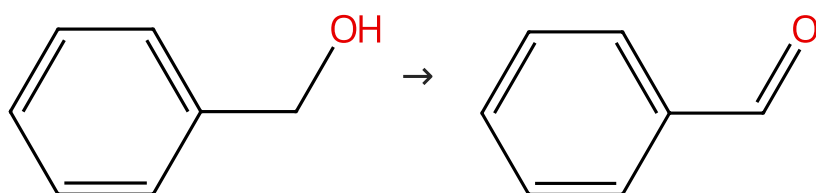
By: Ley, Steven V.; et al

Journal of the Chemical Society, Perkin Transactions 1: Organic and Bio-Organic Chemistry (1999), (6), 669-672.

- 1.1 **Reagents:** Iodobenzene diacetate (polymer-supported)  
**Solvents:** Dichloromethane

Scheme 40 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (149)

Suppliers (69)

31-480-CAS-123227

Steps: 1 Yield: 100%

**Catalytic oxidations of alcohols to carbonyl compounds by oxygen under solvent-free and transition-metal-free conditions**

By: Herrerias, Clara I.; et al

Tetrahedron Letters (2006), 47(1), 13-17.

- 1.1 **Reagents:** Oxygen  
**Catalysts:** Tempo, Iodobenzene diacetate, Potassium nitrite;  
 15 h, 80 °C

31-480-CAS-9446

Steps: 1 Yield: 100%

**Polymer-supported hypervalent iodine reagents in 'clean' organic synthesis with potential application in combinatorial chemistry**

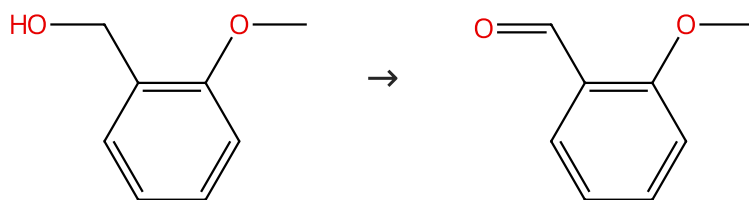
By: Ley, Steven V.; et al

Journal of the Chemical Society, Perkin Transactions 1: Organic and Bio-Organic Chemistry (1999), (6), 669-672.

- 1.1 **Reagents:** Iodobenzene diacetate (polymer-supported)  
**Solvents:** Dichloromethane

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (96)

31-480-CAS-8557413

Steps: 1 Yield: 100%

**Polymer-supported hypervalent iodine reagents in 'clean' organic synthesis with potential application in combinatorial chemistry**

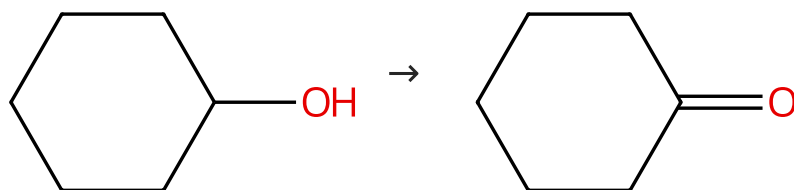
By: Ley, Steven V.; et al

Journal of the Chemical Society, Perkin Transactions 1: Organic and Bio-Organic Chemistry (1999), (6), 669-672.

1.1 **Reagents:** Iodobenzene diacetate (polymer-supported)  
**Solvents:** Dichloromethane

Scheme 42 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (81)

Suppliers (102)

31-480-CAS-9265364

Steps: 1 Yield: 100%

**Clean and highly selective oxidation of alcohols by the Ph I(OAc)<sub>2</sub>/Mn(TPP)CN/Im catalytic system**

By: Karimipour, Gholam Reza; et al

Journal of Chemical Research (2007), (4), 252-256.

1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Imidazole, (*SP*-5-31)-(Cyano-κC)[5,10,15,20-tetraphenyl-21*H*,23*H*-porphinato(2-)-κN<sup>21</sup>,κN<sup>22</sup>,κN<sup>23</sup>,κN<sup>24</sup>] manganese  
**Solvents:** Dichloromethane; 30 min, rt

31-480-CAS-6685107

Steps: 1 Yield: 100%

**A mild and efficient oxidation of alcohols in water**

By: Xu, Chuan Zhi; et al

Chinese Chemical Letters (2004), 15(10), 1149-1152.

1.1 **Reagents:** Iodobenzene diacetate  
**Catalysts:** Tetrabutylammonium bromide  
**Solvents:** Water; rt  
 1.2 **Solvents:** Water; 1 h, rt